EXPERIMENT NO. 9

FLOWCHART TO FIND THE LENGTH OF A STRING USING RAPTOR

Tools Used: Raptor	**Course Outcome:** CO1	**Date:** 28/10/26

PROBLEM STATEMENT

The string is a sequence of characters placed in double quotes (" "). Performing different operations on string data is called String Handling. Strings are immutable. Whenever a change to a String is made, an entirely new String is created. If we want to store a group of characters we can use a char array. String provides various methods to perform different operations on strings. Using Raptor draw a flowchart to display the total number of characters in the string and returns it.

AIM

To draw and execute a flowchart in RAPTOR that reads a string from the user, calculates the total number of characters in the string and displays it.

SOFTWARE / TOOLS REQUIRED

- RAPTOR (Rapid Algorithmic Prototyping Tool for Ordered Reasoning)
- Windows PC

THEORY

A string is a sequence of characters enclosed in double quotes. Strings are immutable – any change creates a new string. String handling means performing operations such as finding the length, concatenation, comparison and searching.

RAPTOR is a flowchart-based programming environment. Programs are built by connecting symbols, and the flowchart can be executed directly, which helps students visualise algorithms.

RAPTOR Symbol	Purpose
Assignment	Stores a value in a variable (x ← expression)
Call	Calls a procedure
Input	Reads a value from the user
Output	Displays a value
Selection	Decision (if–else)
Loop	Repetition until a condition is true

The built-in function **Length_Of(str)** returns the number of characters in the string str.

Algorithm: 1. Start 2. Read str 3. length ← Length_Of(str) 4. Display length 5. Stop

PROCEDURE

Step 1: Launch RAPTOR and Create a New Flowchart

- Open the RAPTOR application from the desktop or Start menu.

- Select File → New to create a new flowchart.
- A blank workspace with Start and End symbols will be displayed.

Step 2: Accept the String from the User

- Drag an Input symbol between the Start and End symbols.
- Enter a prompt such as "Enter a String:".
- Store the input in a string variable named str.

Step 3: Calculate the Number of Characters

- Drag an Assignment symbol below the Input symbol.
- Create an integer variable named length.
- Assign the length of the entered string using the string length function.
- Example:

length ← Length_Of(str)

Step 4: Display the Character Count

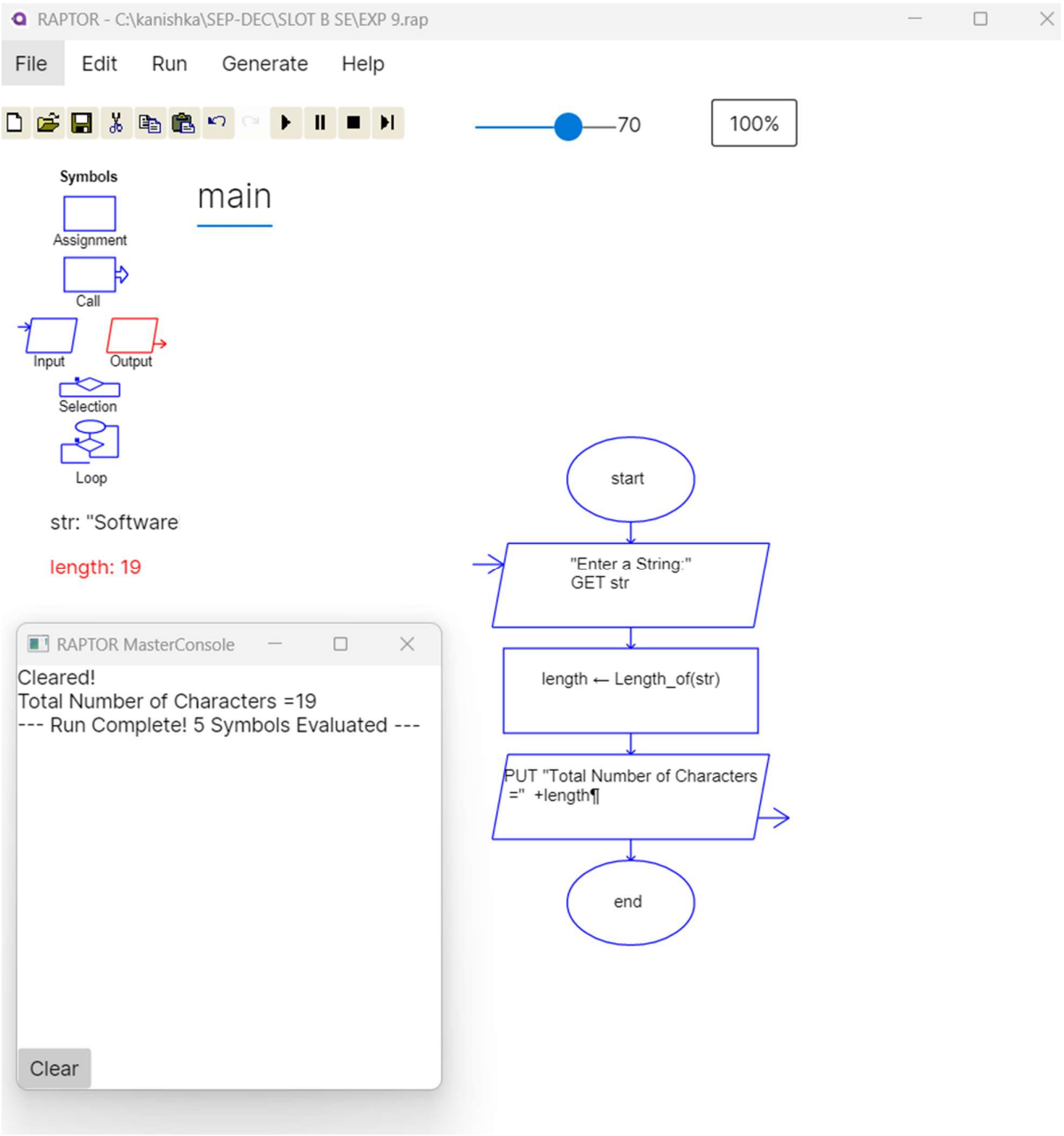
- Drag an Output symbol below the Assignment symbol.
- Display the message "Total Number of Characters =" along with the value stored in the length variable.

Step 5: Connect and Validate the Flowchart

- Connect the symbols in the sequence Start → Input → Assignment → Output → End.
- Check that all symbols are connected properly and the flowchart logic is correct.

Step 6: Execute the Flowchart

- Click the Run button to execute the flowchart.
- Enter a sample string when prompted.
- Verify that the output displays the correct total number of characters in the entered string.



EXPECTED OUTPUT

- Input: Enter a String: SoftwareEngineering
- Output: Total Number of Characters = 19

RESULT

Thus the flowchart to find the length of a given string was drawn, executed and verified using RAPTOR.